

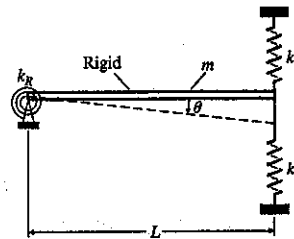
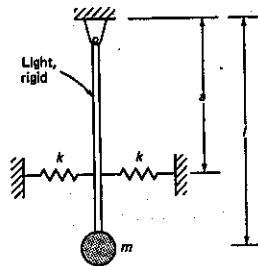
EGM 5421: Structural Dynamics - EXAM I

Instructions:

- (1) The test is open book and open notes. You may work on these problems at your home or any convenient place.
- (2) You may not discuss this test with anyone (from your class or outside). If you are unclear about a problem, you may contact me via email. **I will answer questions received in email by 5pm Friday, February 28, 2025.**
- (3) The test should be turned in by 11:50 p.m. on Sunday, March 2, 2025.

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1. For the systems shown below, write the equations of motion. Assume small deformations.



2. A generator is placed on a base supported by spring-dampers. Spring stiffness is 80kips/inch and the damping ratio is estimated to be 0.2. The weight of the system is 2240lb. At an operating speed of 2000 rpm, the maximum amplitude of the vertical motion is 0.0025inch. Find the force transmitted to the foundation, assuming that the motion is simple harmonic.
3. A cellular telecommunication tower is modeled as an SDOF. To identify the properties, a lateral force of 200 lb is applied at the top of the tower resulting in a horizontal displacement of 1 inch. A cable attached to the top of the tower is suddenly severed resulting in a free vibration. At the end of 2 complete vibration cycles, the time is 1.25 sec and the amplitude is 0.64 inch. Find effective stiffness, effective mass, undamped natural frequency and damping ratio for the SDOF system.
4. A frictionally damped SDOF system is undergoing free vibration after an initial displacement of 6-inch. The weight of the system is 1 kip, stiffness is 2 kips/inch and the friction coefficient is 0.15. Determine the displacement amplitude after 4 cycles and the number of cycles before the system comes to rest. Initial velocity may be assumed to be zero.
5. A simply supported W24x55 steel beam of span 40ft supports a weight of 20kips at its mid span. Upon free vibration, the system exhibits an amplitude decay of 0.75% per cycle. Find the equivalent viscous damping coefficient for the beam. Young's modulus of beam is 29000 ksi.

Q.1 For the systems shown below write equations of motion.

Assume small deformations.

Solution:-

(a) $f_I + f_s + f_{\text{gravity}} = 0$

$$\rightarrow I\ddot{\theta} + k_s \theta + k_{\text{gravity}} \cdot \theta = 0$$

$$\rightarrow I = mL^2$$

One spring, $k_s = k_a$.

Two springs, $k_s = 2k_a$

$$\rightarrow k_{\text{gravity}} = mgL$$

$$\rightarrow mL^2\ddot{\theta} + 2ka^2\theta + mgL\theta = 0$$

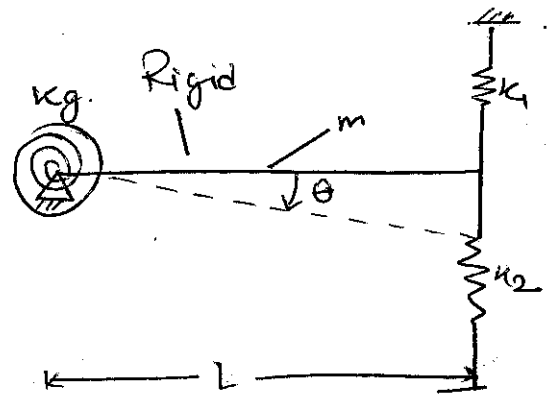
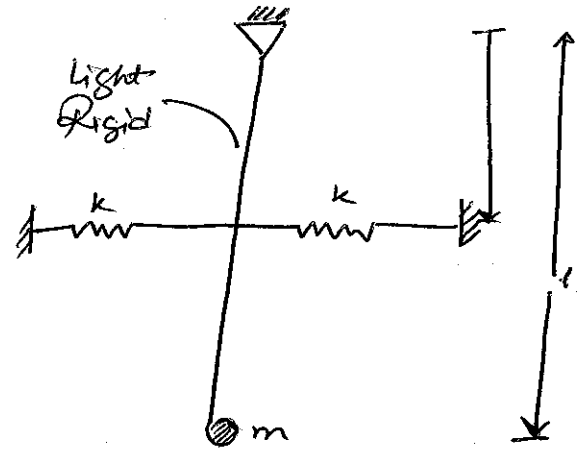
(b) $f_I + f_R + f_{sL} = 0$

$$\rightarrow I\ddot{\theta} + k_R\theta + k_s \cdot L^2\theta = 0$$

$$\rightarrow I = \frac{mL^2}{3}$$

$$k_s = k_1 + k_2$$

$$\rightarrow \frac{mL^2}{3}\ddot{\theta} + k_R\theta + (k_1 + k_2) \cdot L^2\theta = 0$$



Q12

Solution:

Given,

$$k = 80 \text{ kips/in.}$$

$$\xi = 0.2$$

$$\omega = 2,240 \text{ lb.}$$

$$u_0 = 0.0025 \text{ in.}$$

$$\omega = 2,000 \text{ rpm} * \frac{2\pi}{60} = 209.44 \text{ rad/s.}$$

$$(f^+)_0 = p_0 * TR$$

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{80,000 \text{ lb/in.}}{\frac{2,240 \text{ lb}}{386 \text{ in/s}^2}}} = 117.41 \text{ rad/s.}$$

$$\beta = \frac{\omega}{\omega_n} = \frac{209.44}{117.41} = 1.78$$

$$TR = \frac{\sqrt{1 + (2\xi\beta)^2}}{\sqrt{(1-\beta^2)^2 + (2\xi\beta)^2}} = \frac{\sqrt{1 + (2 * 0.2 * 1.78)^2}}{\sqrt{(1-1.78^2)^2 + (2 * 0.2 * 1.78)^2}} = 0.563.$$

$$u_0 = (ust)_0 R_d$$

$$R_d = \frac{1}{\sqrt{(1-\beta^2)^2 + (2\xi\beta)^2}} = \frac{1}{\sqrt{(1-1.78^2)^2 + (2 * 0.2 * 1.78)^2}} = 0.458$$

$$(u_{st})_0 = \frac{u_c}{R_d} = \frac{0.0025}{0.458} = 0.00546 \text{ in.}$$

$$P_0 = (u_{st})_0 * K = 0.00546 * 80,000 = 436.61 \text{ lb.}$$

$$(f^+)_0 = 436.61 * 0.563 = 245.69 \text{ lb} \sim \boxed{246 \text{ lb}}$$

Q:3

Solution:-

Given, $F = 200 \text{ lb}$

$$u_0 = 1 \text{ in}$$

2 cycles in 1.25 s.

$$u_2 = 0.64 \text{ in.}$$

(a) Effective Stiffness: $k = \frac{F}{u} = \frac{200}{1} = \boxed{200 \text{ lb/in}}$

(b) Undamped Natural Frequency: $T_n = \frac{1.25}{2} = 0.625 \text{ sec.}$

$$\omega_n = \frac{2\pi}{T_n} = \frac{2\pi}{0.625} = \boxed{10.05 \text{ rad/sec.}}$$

(c) Effective Mass

$$m = \frac{k}{\omega_n^2} = \frac{200}{10.05^2} = \boxed{1.98 \text{ lb-sec}^2/\text{in}}$$

(d) Damping Ratio

$$\begin{aligned} \xi &= \frac{1}{2\pi} \ln \frac{u_i}{u_i + j} = \left(\frac{1}{2\pi * 2} \right) \ln \left(\frac{1}{0.64} \right) \\ &= 0.036 \sim \boxed{3.6\%} \end{aligned}$$

Q. 4

Solution:

Given, $u_0 = 6 \text{ in.}$

$\omega = 1 \text{ kip.}$

$K = 2 \text{ kips/in.}$

$\mu = 0.15$

$\dot{u}_0 = 0.$

(a) Displacement amplitude after 4 cycles

$P_0 = 100 \text{ kips.}$

$T = 0.30 \text{ sec.}$

$$u_F = \frac{4F}{K} = \frac{4\mu\omega}{K} = \frac{4 * 0.15 * 1}{2} = 0.3 \text{ in.}$$

$$u_4 = u_0 - 4u_F = 6 - (4 * 0.3) = \boxed{4.8 \text{ in.}}$$

(b) Number of cycles before the system comes to rest

$$0 = 6 - (0.3x)$$

$$\Rightarrow x = \frac{6}{0.3} = \boxed{20 \text{ cycles}}$$

Q.5

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Solution

Given,

$$I = 24 * 55$$

$$L = 40 \text{ ft}$$

$$W = 20 \text{ kips at mid span}$$

$$E = 29,000 \text{ ksi}$$

0.75% decay of amplitude per cycle.

$$K = \frac{48EI}{L^3} = \frac{48 * 29,000,000 * 1,350}{(40 * 12)^3} = 16,992.19 \text{ lb/in}$$

$$m_{\text{load}} = \frac{\text{load}}{g} = \frac{20,000}{386} = 51.81 \text{ lb}$$

$$m_{\text{beam}} = \frac{W}{g} = \frac{55 * 40}{386} = 5.70 \text{ lb}$$

$$\omega_n = \sqrt{\frac{K}{m_{\text{eff}}}} = \sqrt{\frac{16,992.19}{51.81 + 5.70}} = 17.19 \text{ rad/sec.}$$

for very low damping,

$$\xi = \frac{u_1 - u_2}{2\pi u_2} = \frac{0.0075}{2\pi} = 1.193 * 10^{-3}$$

$$c = \xi * (2 * m_{\text{eff}} * \omega_n) = 1.193 * 10^{-3} * 2 * 57.51 * 17.19$$
$$= \boxed{2.36 \text{ lb-s/in}}$$